

PHASE ONE PROJECT

Moringa School-Data Science

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Title:

Aviation Accident Analysis

Year Range: 1962–2023

OVERVIEW

Main objective:

Aims to deliver data-driven insights that would provide insights in efforts to improve air safety

Specific objectives:

- To show the most common causes of aviation accidents
- To show how accident rates vary by aircraft type and location
- To understand the role of weather in aviation accident

Data Source:

A csv file from National Transportation Safety Board (NTSB) with accident reports ranging 1962-2023

Business Understanding:

Purpose of this analysis is based on understanding aviation accident trends to improve safety and operations.

DATA UNDERSTANDING

Data Composition:

- Accident date, location, aircraft type, fatalities and contributing factors
- Missing values handled using imputation techniques.
- **Comments on the data**
 - As part of data cleaning, addressed duplicates and handled missing values
 - Derived insights from temporal trends.

DATA ANALYSIS

Some of the analysis I have managed to analyse and make visualizations included:

- ❖ Accidents trend over time
- ❖ Severity of accidents
- ❖ Phase of flights where most accidents take place
- ❖ Fatal and non fatal accidents by weather
- ❖ Number of accidents as influenced by weather
- ❖ Categorization of accidents based on purpose of flights
- ❖ The number of accidents for every make of aircrafts

Recommendation

- The makes recording highest number of accidents be assessed to rule out technical malfunction and so that safety measures on the crafts can be checked and improved
- Avoid flying in bad weather conditions
- The data should have included fatalities caused by pilots; maybe categorized in terms of pilot skill level, level of rest before the flight and number of trips in the same route. This would make the accident analysis more comprehensive
- It was not possible to tell the most common cause of accident using this dataset

Next Step:

- Would be great to do a predictive model for accident risk assessments and monitoring
- Be able to get real time feedback and insights on the dashboard



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